

Lecture 21: Singular Value Decomposition (second part)

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[Based on slides for „Decoding the Matrix“]



WEEK

ELEVEN



Agenda

Chapter 11

Lecture 21:

Closest dimension-k vector space
Singular Value Decomposition (SVD)
Transpose of the SVD
Solving least square problem





Lecture 20 - Recap

Previous

QR factorization

Tool

Trolley solution

Previous

Pythagorean theorem

Concept & Definition

Low rank matrix

Frobenius norm

Trolley line problem

Best rank-one approximation

First Singular value, Right Singular vector, Left Singular vector

Data centering

Closest low dimension-k vector space (1)





Closest low dimension-k vector space (2)

```
def find_right_singular_vectors(A):
    for  $i = 1, 2, \dots, \min\{m, n\}$ 
         $\mathbf{v}_i = \arg \max\{ \|A\mathbf{v}\| : \|\mathbf{v}\| = 1, \mathbf{v} \text{ is orthog. to } \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{i-1} \}$ 
         $\sigma_i = \|A\mathbf{v}_i\|$ 
    until  $A\mathbf{v} = \mathbf{0}$  for every vector  $\mathbf{v}$  orthogonal
        to  $\mathbf{v}_1, \dots, \mathbf{v}_i$ 
    let  $r$  be the final value of the loop variable  $i$ .
    return  $[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r]$ 
```

Theorem: Let A be an $m \times n$ matrix, and let $\mathbf{a}_1, \dots, \mathbf{a}_m$ be its rows. Let $\mathbf{v}_1, \dots, \mathbf{v}_r$ be its right singular vectors, and let $\sigma_1, \dots, \sigma_r$ be its singular values. For $k = 1, 2, \dots, r$, $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the k -dimensional vector space $\mathcal{V}$ that minimizes $(\text{distance from } \mathbf{a}_1 \text{ to } \mathcal{V})^2 + \dots + (\text{distance from } \mathbf{a}_m \text{ to } \mathcal{V})^2$



Deriving the singular value decomposition (SVD) (1)

$$\mathbf{a}_i = \sigma_{i1} \mathbf{v}_1 + \cdots + \sigma_{ir} \mathbf{v}_r$$

$$\mathbf{a}_i = \langle \mathbf{a}_i, \mathbf{v}_1 \rangle \mathbf{v}_1 + \cdots + \langle \mathbf{a}_i, \mathbf{v}_r \rangle \mathbf{v}_r$$

$$\mathbf{a}_i = \begin{bmatrix} \langle \mathbf{a}_i, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_i, \mathbf{v}_r \rangle \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} = \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_1, \mathbf{v}_r \rangle \\ \langle \mathbf{a}_2, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_2, \mathbf{v}_r \rangle \\ \vdots & & \vdots \\ \langle \mathbf{a}_m, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_m, \mathbf{v}_r \rangle \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix}$$



Deriving the singular value decomposition (SVD) (2)

$$\begin{bmatrix} \hline \mathbf{a}_1^T \\ \hline \mathbf{a}_2^T \\ \hline \vdots \\ \hline \mathbf{a}_m^T \end{bmatrix} = \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_1, \mathbf{v}_r \rangle \\ \langle \mathbf{a}_2, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_2, \mathbf{v}_r \rangle \\ \vdots & & \vdots \\ \langle \mathbf{a}_m, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_m, \mathbf{v}_r \rangle \end{bmatrix} \begin{bmatrix} \hline \mathbf{v}_1^T \\ \hline \vdots \\ \hline \mathbf{v}_r^T \end{bmatrix}$$

$$\begin{bmatrix} \langle \mathbf{a}_1, \mathbf{v}_j \rangle \\ \langle \mathbf{a}_2, \mathbf{v}_j \rangle \\ \vdots \\ \langle \mathbf{a}_m, \mathbf{v}_j \rangle \end{bmatrix} = A \mathbf{v}_j$$

Definition 11.3.7: The vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r$ such that $\sigma_j \mathbf{u}_j = A \mathbf{v}_j$ are the *left singular vectors* of A .



Deriving the singular value decomposition (SVD) (3)

$$\begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} = \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_1, \mathbf{v}_r \rangle \\ \langle \mathbf{a}_2, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_2, \mathbf{v}_r \rangle \\ \vdots & & \vdots \\ \langle \mathbf{a}_m, \mathbf{v}_1 \rangle & \cdots & \langle \mathbf{a}_m, \mathbf{v}_r \rangle \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix}$$

Using the definition of left singular vectors, we substitute $\sigma_j \mathbf{u}_j$ for $A\mathbf{v}_j$, resulting in the equation

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} \sigma_1 \mathbf{u}_1 & \cdots & \sigma_r \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix}$$

Finally, we separate out $\sigma_1, \dots, \sigma_r$ into a diagonal matrix, obtaining the equation

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix} \quad (11.12)$$



Existence of singular value decomposition (SVD)

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \hline \mathbf{v}_1^T \\ \vdots \\ \hline \mathbf{v}_r^T \end{bmatrix}$$



Example



SVD of the transpose

$$\begin{bmatrix} A \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix}$$



Best rank- k approximation in term of the SVD

Theorem 11.3.12: Let A be an $m \times n$ matrix, and let $\mathbf{a}_1, \dots, \mathbf{a}_m$ be its rows. Let $\mathbf{v}_1, \dots, \mathbf{v}_r$ be its right singular vectors, and let $\sigma_1, \dots, \sigma_r$ be its singular values. For any positive integer $k \leq r$, $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the k -dimensional vector space $\mathcal{V}$ that minimizes

$$(\text{distance from } \mathbf{a}_1 \text{ to } \mathcal{V})^2 + \dots + (\text{distance from } \mathbf{a}_m \text{ to } \mathcal{V})^2$$

and the minimum sum of squared distances is $\|A\|_F^2 - \sigma_1^2 - \sigma_2^2 - \dots - \sigma_k^2$.

$$\|A\|_F^2 - \|A\mathbf{v}_1\|^2 - \|A\mathbf{v}_2\|^2 - \dots - \|A\mathbf{v}_k\|^2$$

Theorem 11.3.13: For $k \leq \text{rank } A$, the best rank-at-most- k approximation to A is

$$\tilde{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T$$

for which $\|A - \tilde{A}\|_F^2 = \|A\|_F^2 - \sigma_1^2 - \sigma_2^2 - \dots - \sigma_k^2$.



Best rank- k approximation in term of the SVD (2)

Theorem 11.3.13: For $k \leq \text{rank } A$, the best rank-at-most- k approximation to A is

$$\tilde{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T$$

for which $\|A - \tilde{A}\|_F^2 = \|A\|_F^2 - \sigma_1^2 - \sigma_2^2 - \cdots - \sigma_k^2$.



Politics revisited: best rank-one approximation

We center the voting data, and find the closest one-dimensional vector space $\text{Span}\{\mathbf{v}_1\}$.

Now projection along $\mathbf{v}$ gives better spread.



Which of the senators to the left of the origin are Republican?

```
>>> {r for r in senators if is_neg[r] and is_Repub[r]}  
{'Collins', 'Snowe', 'Chafee'}
```

Similarly, only three of the senators to the right of the origin are Democrat.



Example: extended senators low rank matrix

First center the data. Then find first two right singular vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Projecting onto these gives two coordinates.

To find singular vectors,

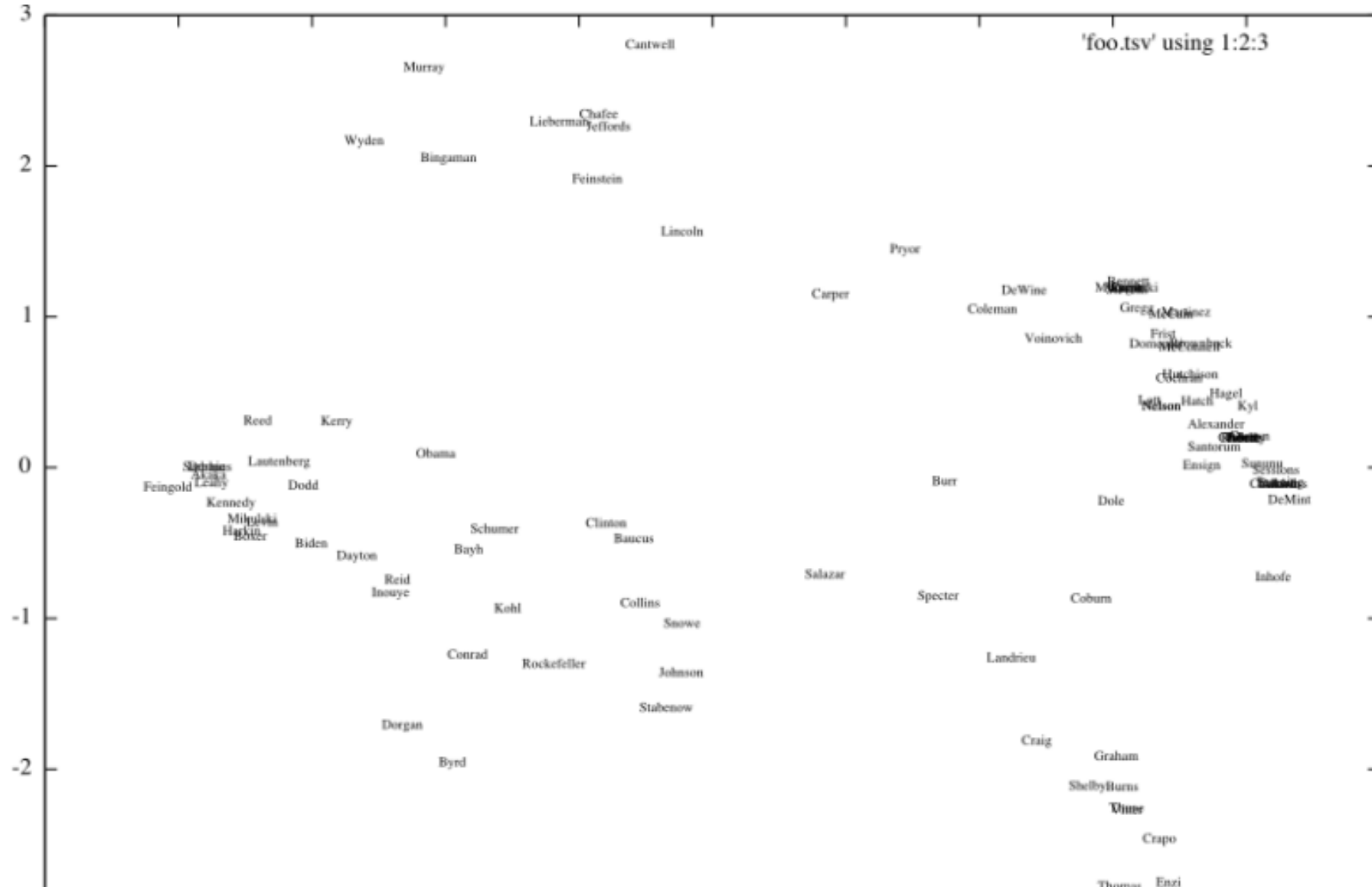
- ▶ make a matrix A whose rows are the centered versions of vectors
- ▶ find SVD of A using `svd` module.

```
>>> U, Sigma, V = svd.factor(A)
```

- ▶ first two columns of V are first two right singular vectors.



Example senators with rank-2 approximation





Uses of SVD

The most famous use of SVD is in principal components analysis and its cousins.

However, SVD is useful for more prosaic problems:

- ▶ Computing rank: rank is the number of singular values above some small specified tolerance.
- ▶ Useful in computing orthonormal bases of $\text{Null } A$ and $\text{Col } A$.
- ▶ least-squares: unlike QR decomposition, SVD can be used even when matrix A does not have linearly independent columns.



Closest k -dimensional affine space

Use the centering technique:

Find the centroid $\bar{\mathbf{a}}$ of the input points $\mathbf{a}_1, \dots, \mathbf{a}_m$, and subtract it from each of the input points. Then find a basis $\mathbf{v}_1, \dots, \mathbf{v}_k$ for the k -dimensional vector space closest to $\mathbf{a}_1 - \bar{\mathbf{a}}, \dots, \mathbf{a}_m - \bar{\mathbf{a}}$. The k -dimensional affine space closest to the original points $\mathbf{a}_1, \dots, \mathbf{a}_m$ is

$$\{\bar{\mathbf{a}} + \mathbf{v} : \mathbf{v} \in \text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_k\}\}$$



SVD and least squares

Algorithm for finding minimizer of $\|\mathbf{b} - A\mathbf{x}\|$:

Find singular value decomposition (U, Σ, V) of A
return $V\Sigma^{-1}U^T\mathbf{b}$



Lecture 21 - Recap

Previous

Best rank-one approximation

First Singular value, Right Singular vector, Left Singular vector

Singular Value Decomposition (SVD)

Solving $Ax = b$

Best rank-k approximation

Concept & Definition



Thank You For Your Attention!





Theoretical exercise

Problem: Let A be a matrix over the reals composed of m rows and n columns, $v_1, \dots, v_k$ the orthonormal k -right singular vectors and $u_1, \dots, u_k$, the k -left singular vectors associated to A show that $u_1, \dots, u_k$ are orthonormal vectors.



Applied exercise

Problem: Let be U , V and Σ three matrix over the reals, check that the result of $U\Sigma V^T$ is equal to $(U\Sigma V^T)^T$.

$$U = \begin{bmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$V^T = \begin{bmatrix} -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$